



STAINLESSCARBON

A carbon and energy calculator for Jindal Stainless, calibrated to its FY26 filings, per plant, with the EU carbon-border figure and the rupee value on the same screen

ORGANISATION	Jindal Stainless Limited
COMPETITION	Stainless Spark – Engineering Innovation, Building Futures
PROBLEM STATEMENT	PS-3 · Carbon and Energy Calculator
ROUND	2 · 27 September 2026
PROTOTYPE	Live web tool · 415 automated model tests
TEAM	Ritesh Kumar Singh, Hriday Pratap Singh, Vaibhav Shukla
INSTITUTION	MNIT Jaipur
CONTACT	[roll numbers / email]

“The EU border charge counts the fuel we burn and the alloys we buy, not the electricity we use.”

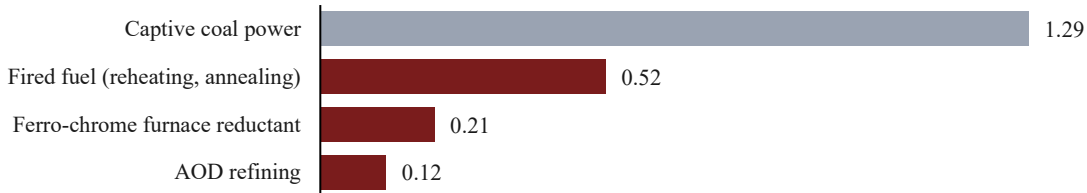
THE PROBLEM AND WHAT WE LEARNED FROM IT

Stainless carbon is mostly alloys and coal power, and the EU border charge counts direct emissions plus bought-in precursors, per plant, not the company number Jindal reports. A useful calculator has to hold both views at once.

The problem, in Jindal’s own numbers

1.76 / 3.03 t CO ₂ e per t crude, FY26: Scope 1+2 / total as disclosed; energy 16.43 GJ/t <i>JSL Integrated Report FY26</i>	2.31 vs 0.74 Scope 1+2 per t crude, Jajpur (captive coal plant, ferro-chrome furnaces) vs Hisar <i>JSL Integrated Report FY26, per plant</i>	1.17 vs 7.14 EU figure per t coil at Jajpur vs the EU default for India, 6.49 + 10% mark-up in 2026 <i>Our model; IR (EU) 2025/2621</i>
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Where Jajpur’s Scope 1 comes from — t CO₂ per t crude steel, **2.14** in total



Grey = outside the EU figure (electricity, even when made on site: 60% of Jajpur Scope 1); red = counted. Captive: 264 MW × load factor 0.70 (our assumption) × 1.045 t/MWh (CEA). Calibrated to Jajpur’s disclosed Scope 1.

Why scrap barely moves Scope 1. Scrap displaces virgin nickel and ferro-chrome, whose carbon sits mostly upstream in Scope 3. +10 pp scrap cuts the total by **0.54** t/t but Scope 1+2 by only **0.17** and the EU figure by **0.15**: the furnaces, the coal plant and the fuel stay.

Why renewables do not move the EU figure. For steel, CBAM counts direct emissions; electricity is indirect whether it is bought or made in the captive plant. +10 pp renewables cuts Scope 1+2 by **0.04** t/t and the EU figure by exactly **0**. Clean power pays in India’s carbon market, not at the EU border.

Four insights that shaped the design

- 1

The alloy chain is Scope 3 — and a CBAM precursor. Nickel is **36%** of the FY26 total and alloys **51%** with ferro-chrome. Bought nickel pig iron adds **0.28** t/t crude to Jajpur’s EU figure; if the supplier stays on EU defaults the figure rises from **1.17** to **1.64** per t coil.
- 2

Captive coal is 60% of Jajpur’s Scope 1 and outside CBAM. It is the largest Scope 1+2 lever and worth nothing at the EU border. **The 2035 finding:** JSL’s **0.99** Scope 1+2 target is reachable only by running the coal plant at about **40%** load (with **39%** clean fuel and **90%** renewables). Leave the coal plant alone and even the theoretical floor is **1.02**.
- 3

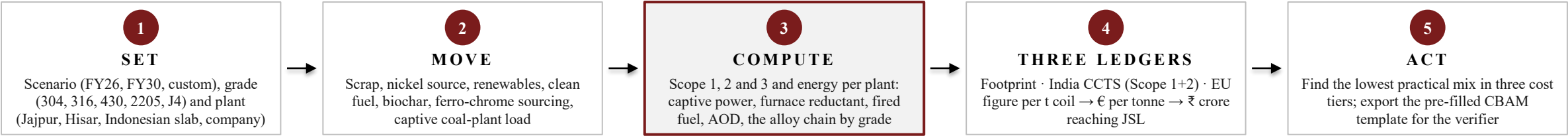
CBAM is filed per installation and per tonne of coil. The company average 1.76 is filed by nobody. Verifiers certify Jajpur **1.17** and Hisar **0.89** per t coil; Indonesian-melted slab rolled in India carries **9.43**.
- 4

Practical and theoretical must be kept apart. Three tiers with disclosed costs: practical today **2.30**, stretch 2030 **1.38**, theoretical floor **0.73** t/t. The last is a bound, not a plan.

PROPOSED SOLUTION — STAINLESSCARBON

03 / 08

Plant, grade and seven levers go in; Scope 1, 2 and 3, energy, the per-installation EU figure and its rupee value come out. Deterministic, calibrated to JSL’s FY26 filings, no generative step anywhere in the numbers.



Every page load re-runs 28 checks against JSL disclosures; 415 model and 110 business-layer tests pass before any number is shown.

What is new

- **Per-installation EU figure with precursors.** Direct carbon minus captive power, plus bought ferro-chrome and nickel pig iron, per t coil; supplier data or EU defaults.
- **Captive coal as a costed lever**, in rankings, optimiser and goal-seek.
- **Three ledgers from one run:** footprint, India CCTS, EU figure → ₹.
- **Indonesian slab route:** 9.43/t coil, €619/t worse than Jajpur.

Why calibrated, not generic

- **Fits FY26 by construction** per plant; FY24 Scope 1 out of sample +0.8%.
- **Every constant tagged:** JSL-assured, regulation, literature, derived, assumption or fitted, with its source.
- **Honesty list published:** the nickel pig iron share is a plug; the coal-plant load is assumed.
- **Generic factors miss the alloy chain** and the captive plant, which is where JSL’s carbon is.

What the user gets

- **Hero numbers:** footprint, energy, ₹ value, with a FY26 → scenario compare strip.
- **Lowest practical mix** in three tiers, every step’s \$/t CO₂ shown.
- **Pre-filled CBAM template** per plant (CSV / JSON).
- **Per-order quote:** € per t coil by grade and plant, before booking.
- **One-page print** for the file.

StainlessCarbon

Carbon and energy calculator for Jindal Stainless steel

SCENARIO: FY26 today, JSL roadmap FY30, Theoretical maximum

STEEL GRADE: 304, 316, 430, 2205, J4

PLANT: Jajpur, Hisar, Indonesia JV slab, Company average

MATERIALS AND FUEL

Recycled scrap in the furnace charge: 70% (move slider)

Clean fuel in furnaces and reheating: 2% (move slider)

WHERE THE POWER COMES FROM: 47% (move slider)

Captive coal plant load: 70% (move slider)

More levers: Model, ferro-chrome, biochar, iron units, EU data rates, EU volume and quality, T per C, coil price

Carbon footprint

3.03 (per tonne of steel)

Energy

16.43 (GJ/t)

What verified data is worth to JSL: ₹139 Cr

FY26 today — your scenario

Carbon footprint, EU: 3.03 (no change)

Energy, GJ/t: 16.43 (no change)

EU figure, GJ/t: 1.17 (no change)

Net to JSL, a year: ₹139 Cr (no change)

Lowest practical mix

Practical today, Switch 2030, Theoretical maximum, Find the lowest practical mix

Where the carbon comes from: Direct 1.52, Indirect 0.24, Bought in alloys & consumables 1.27

The live tool at FY26 defaults: levers, three hero numbers, the compare strip and the lowest-practical-mix button on one screen.

WORKED EXAMPLE — GRADE 304, FY26, COMPANY VIEW

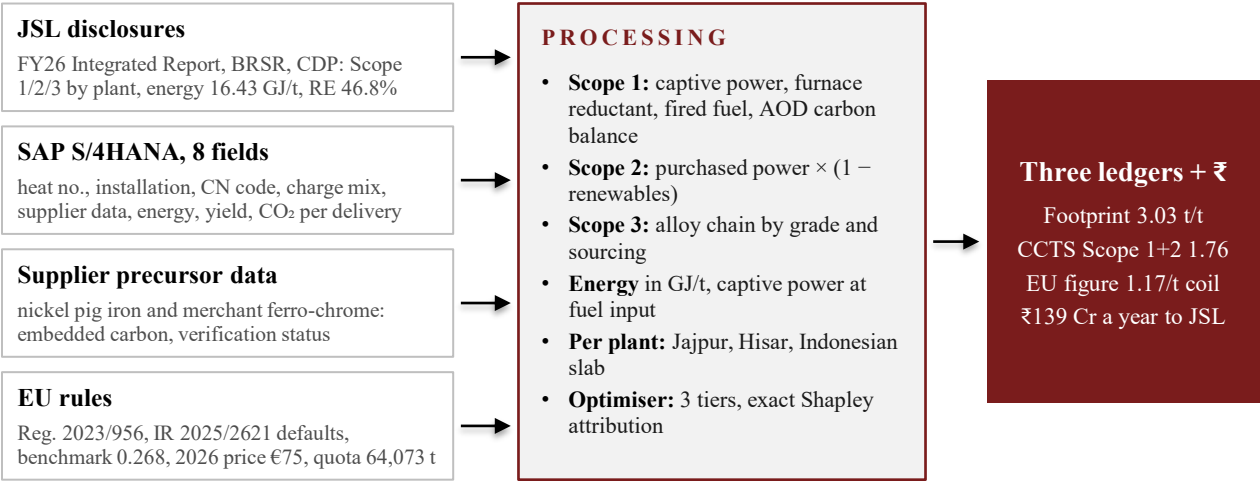
Today: Scope 1 1.52 + Scope 2 0.24 + Scope 3 1.27 = 3.03 t CO₂e per t crude (JSL discloses 3.03); energy 16.43 GJ/t. **Find the lowest practical mix → 2.30 (–24%, about \$27.6 per t steel):** scrap 70 → 75, nickel pig iron share 58 → 45, renewables 47 → 70, clean fuel 2 → 15, biochar 0 → 20, coal-plant load 0.70 → 0.55.

At the EU border (Jajpur): own fuel and process carbon 0.78 + bought precursors 0.28 = 1.05 per t crude = 1.17 per t coil → €68/t in 2026, against €516/t on the India default (certificate €75).

HOW IT WORKS — ARCHITECTURE, DATA AND TECHNOLOGY

One deterministic model file turns four kinds of input into three ledgers and a rupee figure; it runs entirely in the browser, and every constant carries its source and status.

Data pipeline — disclosures and plant data → three ledgers



Today the model runs on public FY26 disclosures; the 90-day pilot connects the SAP and supplier feeds (slide 6).

Where the model still rests on an assumption — and the data that settles it

Item	Status	What settles it
Nickel pig iron share of nickel, 58%	calibration plug	Grade mix by series (200 / 300 / 400), NPI tonnage
Captive coal-plant load factor 0.70	assumption	Captive generation (BRSR Principle 6)
Own ferro-chrome at the site power mix, 4.92 t/t	modelling choice	Coal-plant dispatch when demand falls
Precursor factors, verified and EU-default	to verify	Supplier data; the IR 2025/2621 Annex
Iberjindal as importer of record, 30 kt	to confirm	Iberjindal’s CBAM declarant registration

Technology choices

Layer	Choice	Why
Model	Plain JavaScript module, no dependencies	415 tests incl. 28 checks against JSL filings; one file feeds site, deck, appendix
Interface	Static single page, no backend	Runs offline, on a phone; plant data stays in the browser
Calibration	Per-plant parameters fitted to FY26 filings	Jajpur 2.31, Hisar 0.74 reproduced; FY24 Scope 1 within 0.8%
Optimiser	Rising-cost search, tier caps, Shapley attribution	Cheapest mix per tier; lever shares add up exactly
CBAM engine	Statutory net formula, per t coil, precursor toggle	(E – benchmark × free allocation) × price, per the EC CBAM Q&A, 27 May 2026
Business layer	Quota, pass-through, Iberjindal, verification	Counts only what reaches JSL on in-quota tonnes; 110 tests
Export	CSV / JSON CBAM template	Pre-filled per plant for the verifier and EU declarant
Hosting	Static files	Any web host or the JSL intranet; no server

The eight SAP S/4HANA fields the pilot adds — ₹0.20–0.50 Cr (est.)

- | | |
|---|--|
| 1. Heat / cast number | 5. Precursor supplier, embedded carbon, verification |
| 2. CBAM installation ID | 6. Energy per heat: EAF, AOD, furnace kWh; fuel GJ |
| 3. CN code of the good (7219 / 7220) | 7. Crude → coil yield per route |
| 4. Charge mix per heat, by alloy and scrap type | 8. Embedded emissions + quality flag per delivery |

VALIDATION — CALIBRATED, CHECKED AND CORRECTED

05 / 08

Three kinds of evidence: the model reproduces JSL's filings and FY24 out of sample, every lever is tested under two lenses, and our own Round 1 errors sit next to their fixes.

Reproduces the filings

1.760 FY26 Scope 1+2 (JSL 1.76)	3.031 FY26 total as disclosed (JSL 3.03)
16.43 FY26 energy, GJ/t (JSL 16.43)	2.310 Jajpur Scope 1+2 (JSL 2.31)
0.740 Hisar Scope 1+2 (JSL 0.74)	+0.8% FY24 Scope 1, out of sample (1.716 vs 1.702)

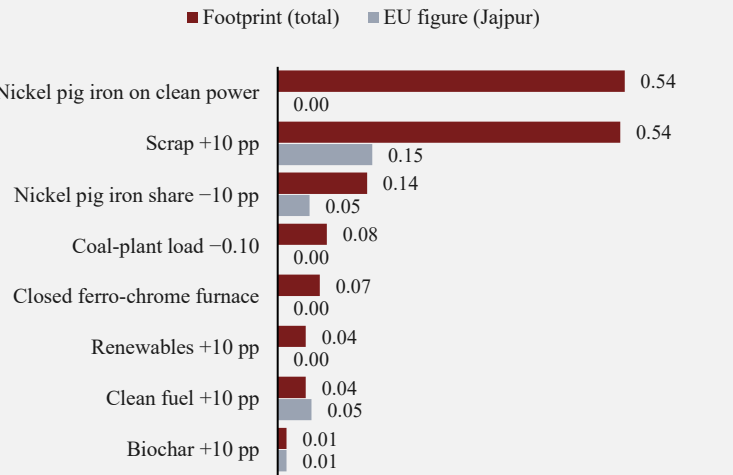
28 of 28 validation checks pass on every page load; 415 model and 110 business tests pass.

Identity, not proof. Seven fitted constants reproduce seven FY26 targets exactly, so the first five matches show consistency, not accuracy. The real tests: FY24 Scope 1 out of sample (+0.8%), and the worldstainless curve at 30% and 50% scrap (6.91 vs 6.80, 3.71 vs 3.70, within ±15%).

Seven fitted constants: crude split, Jajpur and Hisar fired fuel, AOD, purchased power, thermal energy and the nickel pig iron share.

Two lenses on the same levers

Cut in t CO₂e per t, per +10 pp or per switch, FY26. Footprint = company total; EU figure = Jajpur, per t crude.



0 change in the EU figure from renewables or the coal-plant load: both pay in India's carbon market, not at the border

-0.54 scrap slope per 10 pp at JSL, vs -0.8 to -0.9 in the ISSF rule of thumb; the model gives -0.76 on that virgin route

Excluded from both lenses: buying imported ferro-chrome instead of smelting it. It moves 0.44 t from Scope 1+2 into Scope 3 — relocation, never counted as progress.

What we corrected since Round 1

Found in Round 1	Fixed in Round 2
CBAM formula: embedded × phase-in × price (€372–398/t, 2030)	Statutory net formula (E – benchmark × free allocation) × price: €374–448/t in 2026, €736–792 in 2030
Rupee headline: ₹200–290 Cr a year by 2030, buyer-side	₹139 Cr a year net to JSL, in-quota tonnes only
Calibration: S1+2 1.73, total 2.99, company only	1.760 / 3.031, per plant; FY24 out of sample
FY30 path: S1+2 1.35–1.45	S1+2 1.52: the captive coal plant stays on; renewables replace only purchased power
Alloys 60–80% of the footprint	36% nickel, 51% with ferro-chrome at 70% scrap
“No existing calculator does this”	Aperam and Outokumpu publish tools; our edge is the alloy chain and the per-plant EU figure

0.62 430 / 304 footprint ratio at JSL, not “about half”; global ISSF data put it near 0.95.

40 kt EU exports we model, sized to India's 64,073 t quota for all mills; 38.4 kt in quota

Each number above is recomputed by the model rather than restated; the same table opens the research appendix (section 1a), so no superseded figure is left standing unexplained.

IMPLEMENTATION — A 90-DAY JAJPUR PILOT

The model already runs on public data; what remains is connecting Jajpur’s heat data, supplier figures and a verifier — an execution risk, not a research risk.

Who owns what

Owner	Role
Plant heads, Jajpur + Hisar	Signs each plant’s verified emissions report; decides which heats serve EU orders
Head of Sustainability	MRV owner: method, annual report, verifier; files CCTS from the same data
Iberjindal S.L., Spain	Files the EU declaration and surrenders certificates, if importer of record
Procurement	Supplier data clause: nickel pig iron, merchant ferro-chrome, ferro-manganese
EU sales / marketing	Order-level CBAM quote and carbon clause in EU contracts
IT / SAP	The 8 S/4HANA sustainability fields and the delivery output

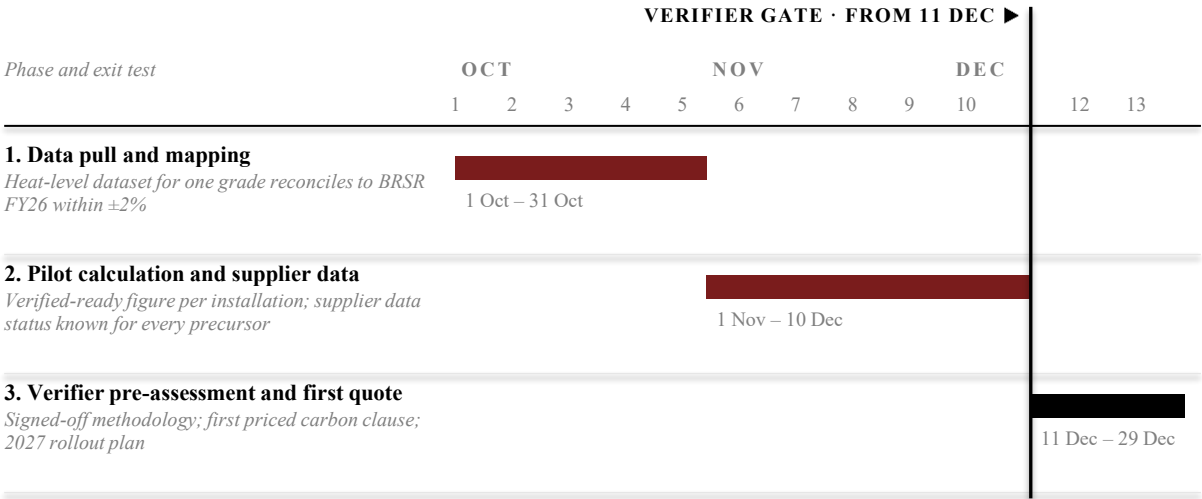
What it costs — ₹ crore, estimates to be replaced by quotes

Line	₹ Cr (est.)	What it covers
Build	0.5–1.1	2 data engineers + 1 MRV/LCA analyst for ~4 months; SAP fields; model hardening
One-off	0.42–0.83	First-year verifier site visits; methodology; CBAM Registry set-up
Running, a year	0.71–1.32	Verifier fees (2 plants), 0.5–1 FTE MRV owner, supplier data programme
First year	1.63–3.25	Build + one-off + one year of running

Payback. Against ₹140 Cr a year captured (₹61 Cr if Iberjindal is not importer of record), the first-year cost is repaid well inside year one (est.). The build alone is repaid if routing keeps ~0.2 kt of Indonesian-slab coil off the EU book.

Scale-up. After the go / no-go on 29 December, the same fields and method extend to Hisar and every EU-bound CN line in 2027. No new software: the running cost stays ₹0.71–1.32 Cr a year for both plants.

90-day plan — 1 October to 29 December 2026



Five data asks that replace our assumptions

1. Captive power generation and load factor (BRSR Principle 6), Jajpur
2. Captive vs merchant ferrochrome split, and ferrochrome sent from Jajpur to Hisar
3. NPI tonnage, the Indonesian JV's power mix, and whether it will provide verified embedded emissions
4. EU export tonnes by installation, CN code and importer of record; quota usage by quarter
5. Grade mix by series (200/300/400) and by installation

Risks. Nickel pig iron supplier won’t verify → Jajpur falls back to 1.64 per t coil, not 1.17 · Iberjindal not importer of record → ₹61 Cr, not ₹139 Cr · heat data off by more than ±2% → phase 2 slips, still inside the 30 Sep 2027 first declaration.

IMPACT — FOOTPRINT, EU BORDER, INDIA, RUPEES

07 / 08

Conservative by construction: we count only in-quota tonnes and only what reaches JSL, not what EU buyers save; out-of-quota tonnes are diverted, and the Iberjindal line is shown both with and without.

₹139 Cr a year

net to JSL from 2026 if Iberjindal is importer of record (**₹61 Cr** if not); **₹246 Cr** by 2030 at the analyst price path

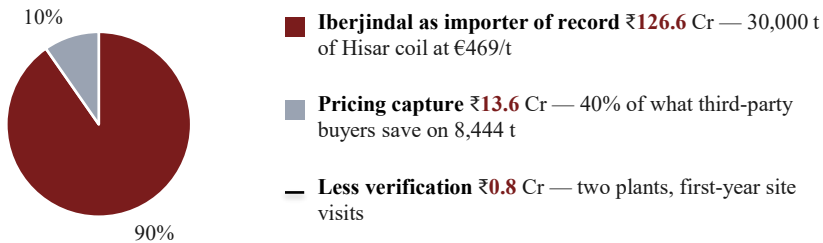
€374–448 per t coil

verified vs India default at Jajpur, 2026; the range is the unsettled default-side benchmark (1.27 → 0.268)

₹56 Cr per 10 kt

what a routing rule saves: Indonesian-slab coil costs **€619/t** more at the border than Jajpur-melted coil

Composition of the ₹139 Cr, 2026



Net = third-party in-quota tonnes × €447.7 × 40% pass-through + Iberjindal tonnes × saving – verification. ₹90/€, certificate €75, 40 kt exports, quota share 60%. All inputs are estimates.

1 Footprint

- **3.03 → 2.30** t CO₂e/t on levers buildable today (–24%, \$27.6/t); **1.38** in stretch 2030; **0.73** theoretical floor.
- **2035 target 0.99 (Scope 1+2)** is reached in stretch: coal plant at **40%** load, **39%** clean fuel, 90% renewables, about \$57.5/t.
- **Energy is reported next to carbon: 16.43** GJ/t today, of which electricity 10.65 and fuels 5.78.

3 India

- **CCTS rules on Scope 1+2 intensity** (draft, 26 Jun 2026), a domestic cost ledger separate from the EU border.
- Renewables and the coal plant pay here: load –0.10 cuts company Scope 1+2 by **0.077**; load 0 with 90% renewables takes it to **0.79**.
- **Captive coal is 55%** of company Scope 1: the lever India’s scheme rewards and the EU border ignores.

2 EU border

- Jajpur **1.17** per t coil vs India default **7.14** → €68 vs €516 a tonne in 2026.
- **Routing rule:** Indonesian-slab coil (**9.43**) is €619/t worse, so EU orders go on Indian-melted heats. India’s quota **64,073** t puts **38.4 of 40 kt** in quota.
- The other **1.6 kt** would pay 50% duty, €1,500/t on a €3,000 coil — more than the carbon saving — so they are diverted.

4 Customers

- **Per-order quote:** the € CBAM cost per t coil by grade and plant, before the order is booked.
- **Product footprint follows the plant: 3.58** Jajpur / **2.59** Hisar per t coil (A1–A3 est.), plus the pre-filled CBAM template for the buyer’s verifier.
- **One dataset, three filings:** the CBAM communication, the CCTS intensity report and the product footprint come from the same heat data.

ASSUMPTIONS, SENSITIVITIES AND REFERENCES

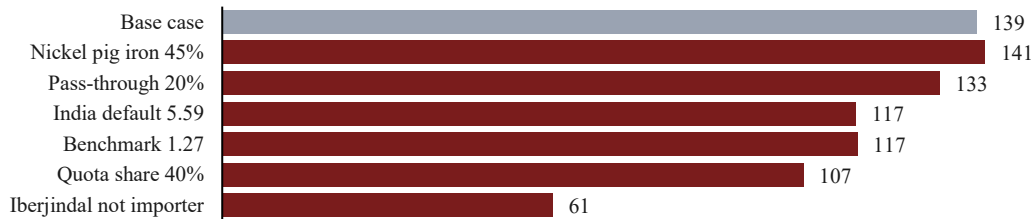
Each input a CFO would push on, moved one at a time: the net to JSL stays between ₹61 and ₹141 crore, and Iberjindal's status matters most. Every source is public.

Sensitivity — net to JSL, ₹ crore a year (base 139)

Assumption	We use	If instead	Net ₹ Cr	Status
Nickel pig iron share	58%	45% / 70%	141 / 138	calibration plug
Coal-plant load, refitted	0.70	0.60 / 0.80	139 / 140; EU figure 1.24 / 0.87 vs 1.05 per t crude	assumption
Default-side benchmark	0.268	1.27	117	not yet published
India CBAM default, base	6.49	5.59	117	regulation, to verify
Iberjindal importer of record	yes, 30 kt	no	61	to confirm
JSL share of India quota	60%	40%	107	assumption

Refit = the seven FY26 fits re-solved at a different load; the net moves under 0.3% because Iberjindal coil is Hisar-melted. Pass-through 20% instead of 40%: ₹133 Cr.

The same tests as a picture — net to JSL, ₹ crore a year



LINKS

Live tool <https://grv-io.github.io/sc-346a9f98/website/>
Research appendix (PDF) <https://grv-io.github.io/sc-346a9f98/appendix.pdf>
The tool opens at FY26 defaults; every figure on these slides can be reproduced there.

COMPANY DISCLOSURES

1. Jindal Stainless Ltd — Integrated Annual Report FY2025-26 (Aug 2026): Scope 1 3,074,950 t; per-plant intensity Jajpur 2.31, Hisar 0.74; energy 16.43 GJ/tcs.
2. Jindal Stainless Ltd — BRSR FY2024-25 and FY2025-26: 70.12% scrap; renewables 46.8% of imported power.
3. Jindal Stainless Ltd — CDP Climate Change disclosures and ESG Factsheets, FY22–FY26.
4. Iberjindal S.L. (Lucena, Spain) — company disclosures: ~30 kt/yr processing.

REGULATION

5. Regulation (EU) 2023/956 establishing a carbon border adjustment mechanism (Art. 31 phase-in).
6. Commission Implementing Regulation (EU) 2025/2620 of 16 Dec 2025 — free-allocation adjustment and benchmarks.
7. Commission Implementing Regulation (EU) 2025/2621 of 16 Dec 2025 — default values, corrected by (EU) 2026/1740 of 31 Jul 2026.
8. European Commission — CBAM Q&A (27 May 2026), §3.8, and the CBAM certificate price page.
9. EU steel safeguard, 2026 revision — India stainless-flat quota 64,073 t; 50% out-of-quota duty (regulation number to confirm).
10. MoEFCC / BEE — Carbon Credit Trading Scheme draft for steel (26 Jun 2026).
11. Regulation (EU) 2024/3110 (Construction Products) and (EU) 2024/1781 (Ecodesign) — product GWP and the Digital Product Passport.

METHOD AND DATA

12. worldstainless / ISSF — Stainless steels and CO₂: industry emissions and related data. <https://worldstainless.org/about-stainless/environment/stainless-steels-and-co2-industry-emissions-and-related-data/>
13. Central Electricity Authority — CO₂ Baseline Database v21.0 (grid 0.710 t/MWh). <https://cea.nic.in/cdm-co2-baseline-database/?lang=en>
14. ICDA — Life-cycle assessment of ferrochrome production (2020–2024).
15. Nickel Institute — nickel LCA; published RKEF nickel pig iron LCAs.
16. ISO 14040/14044, ISO 14404 and the GHG Protocol; worldsteel LCI methodology.
17. Ministry of Steel — Greening the Steel Sector in India (2024).
18. Outokumpu — CBAM Q&A (2026): default-side benchmark example 1.27.
19. SMM (29 Jul 2026) — India CN 7219 default reported at 5.59 base.
20. IPCC 2006 Guidelines, Vol. 2 (Energy); BEE PAT / CEA fuel factors. <https://www.ipcc-nggip.iges.or.jp/public/2006gl/vol2.html>

PRIOR ART CHECKED

Aperam — carbon calculator by origin country; Outokumpu — certified per-batch product carbon footprints; Primetals Technologies — CO₂ Calculator; SSAB — EcoUpgraded. None models an Indian mill's alloy chain per installation on the CBAM boundary.